



TRAINING PROGRAMME

AWS Solutions Architect Associate

Design resilient, secure, high-performing and cost-optimised architectures on AWS

36 HOURS

FOR Engineers and architects who design workloads on AWS

PREPARES FOR AWS Certified Solutions Architect – Associate (SAA-C03)



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Overview

This programme teaches participants to design architectures on Amazon Web Services and prepares the AWS Certified Solutions Architect – Associate exam (SAA-C03). They design for resilience across Availability Zones and Regions, build advanced VPC networks and hybrid connectivity, choose compute, storage and database patterns, decouple applications with queues, events and workflows, secure them across accounts with IAM, KMS, WAF and Organizations, and build serverless and container-based microservices. Almost every chapter combines design discussion with a lab, scenario practice closes the content, and the programme ends with a final project: an architecture designed, built and defended in an architecture review.

At a glance

#	PHASE	FOCUS	HOURS
01	Resilient & Connected Architectures	Multi-AZ and multi-Region design, backup, VPC design and hybrid connectivity	6 h
02	Compute, Storage & Data	EC2 and Auto Scaling, S3 and file storage, data transfer, database patterns	6 h
03	Decoupling & Security	SQS, SNS, EventBridge, Step Functions, IAM strategy, KMS, WAF, Organizations	6 h
04	Serverless & Exam Preparation	Lambda, API Gateway, ECS and EKS, Fargate, SAA-C03 scenarios	6 h
05	Final Project	A multi-tier architecture, defended in an architecture review	6 h
	Buffer	Catch-up, revision, extra lab time	6 h
Total			36 h

Key facts

AUDIENCE	Engineers and architects who design workloads on AWS
PREREQUISITES	Core AWS services (course: AWS Cloud Practitioner), or equivalent hands-on use. Core services are not taught again
DURATION	36 h: 30 h of chapters + 6 h buffer
FORMAT	Design discussions and short lectures followed by guided labs; scenario-based exam practice
LAB ENVIRONMENT	One laptop per participant with the AWS CLI; one AWS sandbox account per team, provided by the organiser, with a budget alert; resources destroyed at the end of each lab to control cost
ASSESSMENT	A lab validated at the end of each phase, scenario practice questions and a final architecture review; certificate of completion for participants who complete the labs and the project
OUTCOME	Participants can design and build a multi-tier, multi-AZ architecture on AWS, justify its cost, resilience and security trade-offs, and sit the SAA-C03 exam

PHASE 01

6 H

Resilient & Connected Architectures

Participants design for failure: workloads spread across Availability Zones and Regions, backups and recovery objectives. They then design the networks underneath: VPC layout, peering, Transit Gateway, endpoints and hybrid connectivity. Validated lab: two VPCs joined through a Transit Gateway, with private access to S3 through a gateway endpoint.

#	CHAPTER	CONTENT	FORMAT	HOURS
1.1	Resilient Architectures	Design principles, multi-AZ, multi-Region, backup	Lecture + Lab	3 h
1.2	Advanced Networking	VPC design, peering, Transit Gateway, endpoints, hybrid connectivity	Lecture + Lab	3 h

PHASE 02

6 H

Compute, Storage & Data

Participants choose instance families and scaling strategies, storage classes and lifecycle rules, file systems and data transfer services, then database patterns with replicas, DynamoDB key design and caching. Validated lab: an Aurora cluster with a read replica and an ElastiCache layer, measured before and after caching.

#	CHAPTER	CONTENT	FORMAT	HOURS
2.1	Compute Patterns	EC2 families and purchase options (On-Demand, Savings Plans, Reserved, Spot), Auto Scaling strategies, containers, placement groups	Lecture + Lab	3 h
2.2	Storage & Data Transfer	S3 classes and lifecycle, EFS and FSx, DataSync, Snow family	Lecture + Lab	1.5 h
2.3	Database Patterns	RDS and Aurora replicas, DynamoDB design, caching	Lecture + Lab	1.5 h

PHASE 03

6 H

Decoupling & Security

Participants decouple components so that one failure does not stop the whole application, then secure the architecture across accounts: IAM strategy, encryption with KMS, edge protection with WAF and Shield, and governance with Organizations. Validated lab: an order flow through SQS, SNS and Step Functions that keeps accepting orders while the processing service is down, with encrypted queues.

#	CHAPTER	CONTENT	FORMAT	HOURS
3.1	Decoupling	SQS, SNS, EventBridge, Step Functions	Lecture + Lab	3 h
3.2	Security Architecture	IAM strategy, KMS, WAF, Shield, Organizations	Lecture + Lab	3 h

PHASE 04

6 H

Serverless & Exam Preparation

Participants build microservices without managing servers, with Lambda and API Gateway or containers on ECS and EKS with Fargate, then work through scenario questions across the four SAA-C03 domains. Validated lab: an API Gateway HTTP API that routes one path to Lambda and another to a Fargate service through a VPC link.

#	CHAPTER	CONTENT	FORMAT	HOURS
4.1	Serverless & Microservices	Lambda, API Gateway, ECS and EKS, Fargate	Lecture + Lab	3 h
4.2	SAA-C03 Preparation	Scenario questions across the four domains; review of services not built in labs (Kinesis, Athena, Glue, EMR, migration services)	Lecture + Workshop	3 h

PHASE 05

6 H

Final Project

Teams of two or three design and build an architecture for a written business case, reusing the phase labs, then defend their choices in an architecture review led by the trainer.

#	CHAPTER	CONTENT	FORMAT	HOURS
5.1	Brief & Design	Read the business case, design a multi-tier, multi-AZ architecture, build its core in the sandbox account	Project	3 h
5.2	Cost, Resilience & Security	Cost estimate, failure scenarios, security controls, architecture diagram and decision record	Project	1.5 h
5.3	Architecture Review	Defend the trade-offs in front of the group, teardown; questions and feedback	Project	1.5 h

FINAL PROJECT

Architecture for a Business Case

A multi-tier, multi-AZ architecture for a written business case, built in the sandbox account and defended in an architecture review that weighs cost, resilience and security trade-offs.

STACK AWS: VPC, EC2 or Fargate, Elastic Load Balancing, Auto Scaling, Aurora or DynamoDB, S3, SQS, KMS, WAF, CloudWatch, AWS Pricing Calculator

DELIVERABLE Architecture diagram, decision record, cost estimate, working core, architecture review